# Data Analytics 344
# A1 Memorandum

## Section A

### Question 1

(a) ZB is the largest
(b) Text, images, audio, video, social media

### Question 2

| | | |
|---|---|---|
| Volume | The quantity of generated and stored data | Requires more memory and computational resources; requires data analytics tools to be scaled to large data sets |
| Variety | Data may be of different types, and from different data sources | Data analytics tools need to be adapted to work with different data types;<br>Data transformations are required to get data into a format required by the data analytics tools |
| Velocity | The speed at which data is generated and processed | Requires data analytics tools to work in an online fashion;<br>Data analytics tools should be computationally efficient;<br>Require incremental models |
| Veracity | Refers to data quality | Data quality issues such as missing values, outliers, different scales, noise need to be corrected |

### Question 3

Value: By extracting hidden information/patterns/trends, value is added to data in the form of improved revenue, reduced costs, improved customer experience, etc.

### Question 4

(a) Stationary refers to a fixed set of data; data set does not change over time.
Non-stationary refers to data streams, where data arrives over time.

(b) May result in concept drift, requiring the data model to be adapted over time; requires online analysis, or real-time analytics Any two.

### Question 5

Data warehouses are central repositories of integrated data from different sources, storing enterprise-wide historical data.

Data marts store a subset of the data located in a data warehouse, oriented to a specific business line or team, or single department.

Data lakes are repositories of data stored in its natural raw format, usually as blobs.

**Question 6**

(a) Data analytics is the process of extracting useful insights from raw data; the process of discovering valuable information from very large databases using algorithms that discover hidden patterns in the data.
(b) Visual analytics utilises interactive visual interfaces to data for the exploration of data.
(c) Any of the following: unstructured data, similarity computation is difficult, similar images may look different, semantic queries on images are difficult without any meta data, noise, blurring, lighting, occlusions.

**Question 7**

(a) Diagnostic analytics
(b) Descriptive analytics
(c) Predictive analytics

**Question 8**

(a) Data understanding
(b) Evaluation

## Section B

**Question 1**

Predictive data analytics is the art of building and using models that make predictions based on patterns extracted from historical data.

**Question 2**

(a) Yes, Model 1 is consistent because it correctly predicts each of the instances in the data set.

(b) Model 1 will generalise better to cases not in the data set, because the model is much simpler than model 2. Model 2 is too specific and overfits the data set.

**Question 3**

(a) The inductive bias of a learning algorithm: Two marks for any of the below
    a. is a set of assumptions about what the true function we are trying to model looks like;
    b. defines the set of hypotheses that a learning algorithm considers when it is learning;
    c. refers to the error from incorrect assumptions in the learning algorithm
(b) If the bias is too high, then relevant relations between the features and targets can be missed, that is the model may underfit. In this case accuracy on both the training set and the test set will be bad. If bias is low, and variance is high, then the model may overfit.

### Question 4

Supervised learning have as its objective to learn a functional mapping between a feature space and a target space. For each instance in the training set, a target value is provided. In supervised learning, no target features are provided, and the objective is to discover hidden patterns among the descriptive features.

## Section C

### Question 1

(a) Ordinal: ID, Policy Type
Nominal: Occupation, Pref Channel
Numeric: Age, Motor Value
Binary: Gender
(b) ID feature should be removed. It is nominal and has cardinality equal to the number of instances. Will have no correlation with the target features

### Question 2

(a) Reduces the change of overfitting.
(c) Results in simpler predictive models.

## Section D

### Question 1

(a) The distribution in figure (b) or (e) is an exponential distribution. The large values in the tail indicate values that should be investigated, as they are likely to be outliers.

(b) The distribution in figure (d) is an uniform distribution, which may indicate unique values for the feature. If the cardinality of the feature is the same as the number of instances, then the feature should be removed.

### Question 2

(a) If we set the number of bins to a very low number we may lose a lot of information about the original distribution. The fewer the number of bins, the more the resulting distribution becomes uniform. The more bins, the more similar the distribution of the original continuous feature is to the distribution of the binned representation. However, if f we set the number of bins to a very high number then we might have very few instances in each bin or even end up with empty bins. Binning also reduces noise.
(b) One-hot encoding should be used, as it maintains the order relation among the values of the original representation.

False

### Question 3

(a) Three features have missing values:

| Feature | Motivation | Plan | Motivation |
|---|---|---|---|
| H.R.Diff | 13.03% from data | Replace with mean | Numeric variable, with |

| | | | |
|---|---|---|---|
| | quality table | over all available values | no outliers, and not too many missing values |
| PREV.TACHY | 44.02% from data quality table | Possibly remove the feature | Too many missing values. |
| TACHYCARDIA | 2.01% from data quality table | Remove instances with missing values, so complete case removal; or impute by as most frequent class for nearest K neighbours | Cannot have instances where target feature has missing values; for imputation, there are few few instances, so imputation based on similarity will not add a significant bias |

(b) Three of the features stands out due to irregular cardinality:

| **Feature** | **Motivation** | **Plan** | **Motivation** |
|---|---|---|---|
| AGE | Has a cardinality of 7 which is very low for a numeric feature | Rather convert to an ordinal feature | It may be better to rather refer to 7 age groups |
| GENDER | Has a cardinality of 4, but only 2 is expected | Standardise values by replacing male with m and female with f | There can be only two possible values for gender |
| TACHYCARDIA<br>PREV TACHY | Considering missing values, there are three possible values, while only two are expected | Nothing | The plan for missing values will remove this data quality issue |

(c) Three features have problems with outliers

| **Feature** | **Motivation** | **Plan** | **Motivation** |
|---|---|---|---|
| HEIGHT<br>WEIGHT | Difference between the 1st quartile and the median values and the difference between the minimum and 1st quartile values are quite different; the histogram of the HEIGHT feature values shows significant skew to the left hand side | Remove the instances with the outlier values | Examining the data we see that there are a number of height values that are a couple of orders of magnitude less than the majority of values; such low heights not possible for provided ages |
| BMI | Unreasonable maximum value for BMI indicate outliers are present; massive skew histogram | Investigate if outliers are due to incorrect values for HEIGHT; if so, do nothing; if not, remove | Removal of HEIGHT outliers will address the BMI outlier problem, because BMI is derived from HEIGHT |
| SYS.B.P | Has a maximum value that is not reasonable | Replace with a missing value, and add to data | Only the one outlier, so imputation will result |

| | | | |
|---|---|---|---|
| | for systolic blood pressure, and that is much further beyond the 3rd quartile than the 3rd quartile is beyond the median | quality plan to impute a value | in no loss of data |

(d) No sampling to be done. The small number of instances with missing values for the target feature will, according to the data quality plan, either me removed or imputed. The two class values, true and false, have very equal frequencies, so classes are not imbalanced.

**Question 4**

(a) The histogram for the different levels of the TACHYCARDIA feature are slightly different suggesting that there is a relationship between the DIA.B.P. and TACHYCARDIA features.

(b) It looks like patients with higher diastolic blood pressure values are more likely to suffer with tachycardia than those with lower blood pressure values.

(c) True. The histograms for the the HEIGHT feature split according to the different target levels look very similar. This suggests that there is no strong relationship between the HEIGHT feature and the target feature, TACHYCARDIA .

(d) No, the missing values do not present a problem. Although there are many missing values for the PREV. T ACHY. feature, they are equally prevalent for both values of the target feature. This is illustrated by the even split between the target levels in the bar on the left hand side of the bottom chart.

(e) When a value for PREV. TACHY. is present it is quite predictive of the target feature: previous occurrences of tachycardia are predictive of feature occurrences.

## Section E

**Question 1**

(d) predicting the oil price

**Question 2**

False

**Question 3**

(a) Data points with outliers
(b) Data points with different densities
(d) Data points with non-convex shapes

**Question 4**

Optimal number of clusters is 3
Optimal number of clusters is where the Silhouette coefficient is maximized

**Question 5**

To prevent a feature with large range to dominate a feature with smaller range

**Question 6**

No, they are not well separated.
The histograms of the different clusters with respect the descriptive feature overlaps too much.

**Question 7**

(a) petal length & petal width; the distributions are well-separated for these features
(b) sepal length & sepal width; distributions overlap so does not help at all to discriminate between the two clusters.
(c) pw in [0,0.6] & pl in [1,2]

Values & ranges are approximate

## Section F

**Question 1**

Bird

**Question 2**

(a) 40
(b) 70
(c) 100
(d) 2.42

+ 2 marks for calculations

**Question 3**

(a) Hair = red
(b) Addition of the condition maximizes accuracy.

**Question 4**

(a) true; Rules are extracted to achieve 100% accuracy on training set, and therefore noisy instances will have their own rules
(b) false; Rule association algorithms require categorical features, so all continuous features have to be transformed to categorical features
(c) false; Ripper is applied to each class in turn, starting from the minority class
(d) true; The empty rule covers none of the instances
(e) false; Association rule mining does not assume a target, but discovers trends among all features, incrementally taking each feature as the consequent
(f) false; the coverage threshold should be increased to reduce the number of conditions in the antecedent of rules.